

## Sec 6.3 – Differential Equations and Motion

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

Often we have information about a derivative and want to find the original function. For example, we might know that

$$\frac{dy}{dx} = 4x^2 + \sin x.$$

A **differential equation** is an equation of the form  $\frac{dy}{dx} = f(x)$ . Finding the **general solution** to the differential equation means finding the general antiderivative  $y = F(x) + C$ .

What is the general solution to  $\frac{dy}{dx} = 4x^2 + \sin x$ ?

$$y = \frac{4}{3}x^3 - \cos x + C$$

Suppose we want a specific function as a solution to a differential equation. How do we pick which one?

To pick, we will have to pick a value of  $C$ . This means we need to know about a specific value of the solution. This is called the **initial condition**.

**Example:** Find  $y$  if  $\frac{dy}{dx} = 3x + 5$  and  $y(2) = -1$ .

$y(2) = -1$  is the initial condition and the overall problem is called an **initial value problem**. An initial value problem usually has a unique solution called the **particular solution**.

$$y = \frac{3}{2}x^2 + 5x + C.$$

The notation  $y(2) = -1$  means that the point  $(2, -1)$  satisfies the equation.

$$\text{So } -1 = \frac{3}{2}(2)^2 + 5(2) + C \text{ or } -1 = 16 + C.$$

$$\text{Therefore } C = -17 \text{ and } y = \frac{3}{2}x^2 + 5x - 17.$$

Next we will consider a common application of solving differential equations, which is finding the equations of motion.

It has been known since Galileo that gravity provides a constant downward acceleration,  $g$ . In the most common units, this acceleration is approximately:

$$g = 9.8\text{m/s}^2 \qquad g = 32\text{ft/s}^2.$$

But, acceleration is the derivative of velocity! So if velocity is positive in the upward direction,  $\frac{dv}{dt} = -g$ . And more generally we have:

position/distance

velocity

acceleration

$s(t)$

$$\frac{ds}{dt} = s'(t) = v(t)$$

$$\frac{dv}{dt} = v'(t) = a(t) \qquad \text{btw, } \frac{d^2s}{dt^2} = a(t)$$

**Example:** A ball is dropped from a 20 m platform. When does the ball hit the ground and what is its velocity at that time?

Given this information, we need to find the position function  $s(t)$ , solving for  $t$  when  $s(t) = 0$ , which is when the ball is on the ground. We will then take that value of  $t$  and plugging it into  $v(t)$  to find the velocity when the ball hits the ground.

We'll use meters, since those are the units given in the problem, starting with  $-g = \frac{dv}{dt} = -9.8$

Since  $v(t)$  is the antiderivative of this equation, we take the integral  $v(t) = \int -9.8 dt = -9.8t + C$ .

As we have discussed previously, this represents a family of functions. But we know that the ball was dropped, so the initial velocity was 0, so this is an **initial value problem** (IVP).  $v(0) = -9.8 \cdot 0 + C = 0$ , which means  $C = 0$ .

Since velocity is the derivative of position  $s(t)$ , we now can do another integral to solve the differential equation:

$$\frac{ds}{dt} = v(t) = -9.8t$$

$$s(t) = \int -9.8t \, dt = -\frac{9.8}{2}t^2 + C$$

We have another IVP, in this case the initial height is 20 meters, so  $s(0) = 20$ ,

$$s(0) = 20 = -\frac{9.8}{2}0^2 + C$$

Which gives  $C = 20$ , and our position function is:  $s(t) = -4.9t^2 + 20$

We are ready to solve for when the ball hits the ground, solving for  $t$  when  $s(t) = 0$ ,

$$0 = -4.9t^2 + 20$$

which gives  $t \approx \pm 2.0203$ , and since this is a physical problem,  $t$  has to be positive, and we have  $t = 2.0203$  sec.

At time  $t = 2.0203$  the velocity is  $v(2.0203) = -9.8 \cdot 2.0203 = -19.799$  meters/sec (**don't forget units**).

The negative sign means the ball is going downward, since positive velocity is in the upward direction.

## Partner up

A stone is thrown straight up into the air with a velocity of 30 ft/s from a height of 4 feet. When does the stone hit the ground and what is its velocity at that time?

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We'll use feet, since those are the units given in the problem, starting with  $-g = \frac{dv}{dt} = -32$

Since  $v(t)$  is the antiderivative of this equation, we take the integral  $v(t) = \int -32 dt = -32t + C$ .

Since the ball was thrown, the initial velocity was 30 feet/sec, and we solve the IVP,  $v(0) = 30 = -32 \cdot 0 + C$ , so  $C = 30$  and the velocity function is:  $v(t) = -32t + 30$ .

Since velocity is the derivative of position  $s(t)$ , we now can do another integral to solve the differential equation:

$$\frac{ds}{dt} = v(t) = -32t + 30$$
$$s(t) = \int -32t + 30 dt = -16t^2 + 30t + C$$

We have another IVP, in this case the initial height is 4 feet, so  $s(0) = 4$ ,

$$s(0) = 4 = -16 \cdot 0^2 + 30 \cdot 0 + C$$

Which gives  $C = 4$ , and our position function is:  $s(t) = -16t^2 + 30t + 4$

We are ready to solve for when the ball hits the ground, solving for  $t$  when  $s(t) = 0$ ,

$$0 = -16t^2 + 30t + 4$$

We have two approaches to find  $t$ , (1) the quadratic equation or (2) Sage/CoCalc:

$$(1) t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-30 \pm \sqrt{30^2 - 4(-16) \cdot 4}}{2 \cdot (-16)} = \frac{-30 \pm \sqrt{1156}}{-32}, \text{ which gives } t = -\frac{1}{8} \text{ or } 2.$$

It has to be 2 since this is a physical problem.

(2) here is a screen shot from Sage/CoCalc:

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In [1]: 1 var('t')
        2 solve(-16*t^2+30*t+4==0,t)
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Out[1]: [t == (-1/8), t == 2]
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At this time,  $v(2) = -32 \cdot 2 + 30 = -34$ , so the stone hits the ground after 2 seconds and at -34 ft/second.

In conclusion for Section 6.3,

We have three ways to talk about the same idea:

- Antiderivatives

$f(x)$  has an antiderivative  $F(x)$ , where  $F'(x) = f(x)$

- indefinite integrals

$\int f(x) dx = F(x) + C$ , a whole family of antiderivatives

- differential equations (general solution)

$\frac{dy}{dx} = f(x)$  has a general solution  $y = F(x) + C$

(you can also find a particular solution if you have an initial value.)